1. Food wastage management → Restaurant, customer section for booking (supply) → Khud kitne hain, time ka ho gya → NGO add krenge → normal khana jo edible hai and unko ek jagah krke animals ko khila denge→accidents bachenge.
ML model→

A) Surplus Prediction (Restaurant side)

👉 Predict:

- Kis din kitna food waste hoga?

Model:

- Time Series Forecasting

Core Tech:

- Python
- Prophet / ARIMA / LSTM
- Pandas
- FastAPI (ML service)

Input Features:

- Day of week
- Festival indicator
- Historical sales
- Weather
- Time

C) Food Quality Detection (Advanced)

Food kb bana tha,season,temperature,ingredients(on the basis of inventory)

2. Local business → locality mein kitne logon ko kya zaroorat hai and usko wo pehle se hi store kr lega for competition with the big giants.

Use Case: Demand Forecasting

Predict:

- Next 7 days demand
- Festival spikes
- Weather impact

Models:

Basic:

- Linear Regression
- Random Forest

Advanced:

- Prophet
- LSTM (Deep Learning)

3. Giveaway corner: Raddi bechna hum logon ko kabad waala banna hai.

Predict scrap price

Model:

- Regression model

Tech:

- scikit-learn

🎯 Use Case 2: Image Classification

User photo upload → classify item.

Model:

- CNN
- Pretrained ResNet / MobileNet

Tech:

- TensorFlow / PyTorch
- ONNX for optimization

LOCAL LINKS

Currently we are solving real problems but they seems to be be some random problems . we have to make them structured so that it meets with our project title that is local links , basically our project will be making some kind of connection between the buyer and seller , person in need with a person with surplus .

Some of such ideas will be :

1. Food wastage management :

Basically we will provide a platform where people can just give away their surplus food that they do not want to waste or can sell it at some cost so that they earn some amount instead of throwing it out . This way ye help karega ki food kam se kam waste ho . and we have to make sure ki food ki quality ke hissab se usse kaise dispose krna hai animals and ngos ko dena hai ya fir just usko as a biodegradable waste ki tarah dispose krna hai . And we will allow restaurants to sell their surplus food at low cost so that they may earn something instead of wasting it .

Restaurant, customer section for booking (supply) → Khud kitne hain, time ka ho gya → NGO add krenge → normal khana jo edible hai and unko ek jagah krke animals ko khila denge→accidents bachenge.

Features like :

Dynamic surplus listing
Real-time pickup matching
Expiry countdown alerts
Quality detection using image + temperature logs

Also add:

Carbon footprint saved counter

Impact dashboard

2. Shared Resource Pool (Reduce Overbuying)

🧠 Problem

People buy:

- Drill machines
- Ladders
- Projectors
- Tents , etc

But use them rarely.

💡 Solution

Hyperlocal rent/share platform:

- Peer-to-peer lending
- Verified users
- Deposit system

Smart Feature

- Demand forecasting for seasonal items
- Recommendation engine

This promotes sustainability.

3. Local Skill Exchange & Micro-Work Marketplace

🧠 Problem

Local people:

- Electricians

- Plumbers
- Tutors
- Carpenters
- Part-time helpers

Have no proper digital presence.

💡 Solution

Hyperlocal service marketplace:

- Verified workers
- Ratings
- Nearby matching
- Instant booking

🤖 Smart Matching

Use ML:

- Recommend best service provider
- Predict demand surge (e.g., summers → AC repair spike)

Impact

Empowers local employment.

4. Local Emergency & Blood/Medicine Network

🧠 Problem

People struggle to find:

- Blood donors
- Rare medicines
- Oxygen cylinders
- Ambulance quickly

💡 Solution

Create a **verified hyperlocal emergency network**:

- Filter by locality
- Real-time availability
- Nearby donors
- Medicine store availability

🚀 Advanced Feature

- ML model to predict **high-demand medicines** in certain seasons
- Heatmap of frequent emergency zones

Tech Stack

- Node + Express
- Geo-location APIs
- PostgreSQL
- ML for seasonal demand

👉 This increases social impact significantly.

5. Hyperlocal Apartment-to-Shop Commerce Model

🧠 Problem

Platforms like

- Blinkit
- Zepto
- Swiggy Instamart

are dominating quick commerce.

Result:

- Small kirana stores lose customers
- Vegetable vendors lose regular buyers
- Money flows to big corporations instead of local economy

People inside apartments:

- Prefer convenience
- Don’t want to step out
- But are unaware of nearby shop inventory

💡 Solution

Create a **Hyperlocal Commerce Layer**

Residents can:

- View nearby shops (within 200–500 m)
- See inventory

- Order items
- Track delivery
- Pay online or COD

Shopkeepers:

- Manage stock via simple dashboard
- Accept orders
- Deliver via:
    - Their own staff
    - Local delivery partners
    - Apartment security coordination

Guys here are the things we need to build in our project :

LOCAL LINK :

Tech:

Frontend: Next.js (App Router, client-side for now)

Backend: Node.js + Express (MERN)

DB: MongoDB : *(Using because we are playing around geolocation(latitudes and all) , so mongoDB provides a specific features of geo index that help in search and manipulate location related data .)*

ML: Independent FastAPI service

Overall Architecture

Next.js (Frontend)

↓

Express API Gateway

↓

Backend Service (Modular) -----> ML Microservice (FastAPI)

↓

MongoDB

Expected FRONTEND STRUCTURE (Next.js)

```
app/
├── layout.tsx
├── page.tsx (Landing)
├── login/
├── register/
├── dashboard/
│   ├── resident/
│   ├── shopkeeper/
│   ├── service-provider/
│   ├── ngo/
│
├── food/
│   ├── page.tsx
│   ├── create/
│   ├── my-listings/
│
├── commerce/
│   ├── page.tsx
│   ├── shop/[id]/
│   ├── cart/
│   ├── orders/
```

```
│
├── skills/
│   ├── page.tsx
│   ├── provider/[id]/
│   ├── bookings/
│
├── resources/
│   ├── page.tsx
│   ├── item/[id]/
│   ├── my-items/
│
├── emergency/
│   ├── page.tsx
│   ├── blood/
│   ├── medicine/
│
components/
services/
utils/
```

Pages Required Per Module

Waste food management

Essential Pages

- /food → Browse surplus
- /food/create → Post surplus
- /food/my-listings -> list all by a user
- /food/[id] -> view details
- /food/claim/[id] -> buy some discount food or claim free food

- NGO dashboard and services

APARTMENT COMMERCE (CORE)

Essential Pages

- /commerce -> show all shops in nearby location

- /commerce/shop/[id] -> show the inventory of a shop

- /commerce/cart -> cart for a user , one cart can contain item from one shop only
- /commerce/orders -> order payment cash/ online

- Shop Dashboard -> shops can maintain their inventory , daily sales and other details
- Admin page -> Adding and verifying shops

SKILL EXCHANGE

Pages

- `/skills  -> list all , search , sort and other features`
- `/skills/provider/[id]  ->view services detail by a user ,reviews also`
- `/skills/bookings -> booking a service should include providing your location and other details and some advanced payment.`
- Provider dashboard -> managing active days , prices , service and all .

SHARED RESOURCE POOL

Essential Pages

- `/resources -> list all resources but with ml prediction and search sort features …`
- `/resources/item/[id]  -> view details of a resource`
- `/resources/my-items  -> manage and add a resource`
- `/ resource/book/[id]  -> booking details and depositing safety amount.`

EMERGENCY NETWORK

Essential Pages

- `/emergency`
- `/emergency/blood`

- `/emergency/medicine`

ML SERVICE STRUCTURE (Separate FastAPI)

Endpoints:

POST /predict-demand
POST /predict-food-surplus
POST /recommend-service-provider
POST /predict-medicine-demand

Our Node backend will call the ML API internally.

Backend Service

For backend service we have to make schema for each entity we require , and make routes and models for each of the five services in a modular structure , and will be having a single server for now where we will be handling all routes with proper api endpoints .

Note : Please be very careful while designing each of the models and the routes .

Documentation for *Local Links*

1. Software Requirements Specification (SRS)

Structure of SRS

1. Introduction

- Purpose of project
- Scope
- Definitions
- Overview

2. Overall Description

- Product perspective
- Product functions
- User classes (Resident, Shopkeeper, NGO, Service Provider)
- Operating environment (Web App – MERN)
- Constraints
- Assumptions

3. Functional Requirements (Very Important)

For each module define:

Example:

FR-1: Apartment Commerce

- User can view nearby shops
- User can add items to cart
- User can place order
- Shopkeeper can update order status

FR-2: Food Waste Module

- Restaurant can list surplus food
- NGO can claim surplus
- System auto-expire listing

Write all 5 modules clearly.

4. Non-Functional Requirements

- Performance
- Security (JWT Auth)
- Availability
- Scalability
- Usability
- Reliability

5. External Interface Requirements

- UI
- APIs
- ML service integration

2. System Architecture Diagram :

Show:

```
Frontend (Next.js)
       ↓
Backend (Node + Express)
       ↓
MongoDB
       ↓
ML Microservice (FastAPI)
```

Explain:

- Modular monolithic architecture
- Geo queries
- JWT authentication
- ML API integration

3. Use Case Diagram

4.Sequence Diagrams (2–3 Important Flows)

5. Data Flow Diagram (DFD)

Level 0:

- User → Local Links → DB

Level 1:
Break into:

- Commerce module
- Food module
- Resource module
- Skill module
- Emergency module

6. Security Design Document

Mention:

- JWT authentication
- Password hashing (bcrypt)
- Role-based authorization
- Input validation
- Geo data protection

7 . README.md should be well structured , fully complete in all terms .

Work to be done per person:

1. Login - signup + Landing + Home + Emergency network Service (Utkarsh)

2. Documentation + Shared resource pool (Khushal)

3. Skill Exchange Service + ML (Lavish)

4. Apartment Commerce + Containerisation + Deployment+ Frontend helper everywhere (Mayank)

5. Waste food Management + Backend helper everywhere (Aditya)

Note : Do not stick to it as we will be doing it as our team project so we will be sharing help and doing it together .

Note 2 : You are free to make your selected service as complex and include technology of your choice if you want to flex your individual work so that you do not blame anyone that I got less credits😉.